

PRACTICAL NUMBER 1

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Practical 1.1.1: Calculate Momentum

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum is calculated using the formula:

where:

is the mass of the object (in kilograms).

is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

#Type Content here...

```
mass = float(input())
```

```
velocity = float(input())
```

```
p = mass * velocity
```

```
print(f"{p:.2f}kgm/s")
```

Average time
0.006 s
6.00 ms

Maximum time
0.007 s
7.00 ms

✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 6 ms DEBUG

Expected output	Actual output
5.0	5.0
10.0	10.0
50.00kgm/s	50.00kgm/s

✓ Test case 2 5 ms DEBUG

Expected output	Actual output
2.3	2.3
4.8	4.8
11.04kgm/s	11.04kgm/s

1.1.2. Conditional Calculation Based on the Number of Digits

03:57

Write a Python program that accepts an integer as input. Depending on the number of digits in .

Constraints:

$$1 \leq \leq 999$$

Input Format:

- The input consists of a single integer .

Output Format:

- If is a single-digit number, print its square.
- If is a two-digit number, print its square root (rounded to two decimal places).
- If is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

#write your code here...

```
import math
```

```
n = int(input())
```

```
if n<10:
```

```
    print(n ** 2)
```

```
elif n<100 and n>9:
```

```
    print(f"{math.sqrt(n):.2f}")
```

```
elif n>99 and n<1000:
```

```
    print(f"{math.cbrt(n):.2f}")
```

else:

```
print("Invalid")
```

1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

The screenshot displays the execution results of a Python program. At the top, performance metrics are shown: Average time 0.004 s (4.14 ms) and Maximum time 0.005 s (5.00 ms). Test results indicate that 4 out of 4 shown test cases passed, and 3 out of 3 hidden test cases passed. Below this, four test cases are detailed:

Test Case	Expected output	Actual output
Test case 1	9 81	9 81
Test case 2	166 5.50	166 5.50
Test case 3	24 4.90	24 4.90
Test case 4	3456 Invalid	3456 Invalid

Input Format:

- The first line contains the first date in the format .
- The second line contains the second date in the format .

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

```
from datetime import date
```

```
# Read input dates as strings
```

```
date_str1 = input()
```

```
date_str2 = input()
```

```
# Split the strings by '-' and convert components to integers
```

```
y1, m1, d1 = map(int, date_str1.split('-'))
```

```
y2, m2, d2 = map(int, date_str2.split('-'))
```

```
# Create date objects
```

```
f_date = date(y1, m1, d1)
```

```
l_date = date(y2, m2, d2)
```

```
# Calculate the difference
```

```
delta = l_date - f_date
```

```
# Output the result (number of days)
```

```
print(delta.days)
```

1.1.4. Reverse a Number

05:35

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- **The input is a single integer.**

Output Format:

- **The output prints a single integer, which is the reversed number.**

#write your code here...

```
n = int(input())
```

```
rev = 0
```

```
while n > 0:
```

```
    d = n%10
```

```
    rev = rev*10 + d
```

```
    n = n // 10
```

```
print(rev)
```

The screenshot shows a code execution environment with the following details:

- Performance Metrics:**
 - Average time: 0.008 s (7.60 ms)
 - Maximum time: 0.012 s (12.00 ms)
- Test Results:**
 - 2 out of 2 shown test case(s) passed
 - 3 out of 3 hidden test case(s) passed
- Test Case 1:** Passed in 12 ms. Expected output: 5367, Actual output: 5367. Below the expected output is the number 7635.
- Test Case 2:** Passed in 4 ms.

1.1.5. Multiplication Table

02:23

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

<number> x <multiplier> = <result>

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

#write your code here...

```
n = int(input())
```

```
for i in range(1,11):
```

```
    print(f"{n} x {i} = {n*i}")
```

Average time
0.007 s
6.75 ms

Maximum time
0.010 s
10.00 ms

✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 8 ms

DEBUG

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Expected output	Actual output
8	8
8 · x · 1 · = · 8	8 · x · 1 · = · 8
8 · x · 2 · = · 16	8 · x · 2 · = · 16
8 · x · 3 · = · 24	8 · x · 3 · = · 24
8 · x · 4 · = · 32	8 · x · 4 · = · 32
8 · x · 5 · = · 40	8 · x · 5 · = · 40

1.2.1. Pass or Fail

21:58

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer , the number of courses.
- The second input will be integers representing the marks of the student in each of the courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- **Aggregate Percentage:** Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

```
# write your code here

n_sub = int(input(""))
m_str = input("").split()
marks = []

for i in m_str:
    marks.append(int(i))

percentage = (sum(marks)/(100*n_sub))*100
f = False

for i in marks:
    if i<40:
        f = True
        break

if f:
    print("Fail")
else:
    print(f"Aggregate Percentage: {percentage:.2f}")
    print("Grade:",end=" ")
    if percentage>75:
        print("Distinction")
    elif percentage>=60:
        print("First Division")
    elif percentage>=50:
        print("Second Division")
    else:
```

```
print("Third Division")
```

The screenshot displays a code execution environment with the following details:

- Performance Metrics:**
 - Average time: 0.010 s (10.30 ms)
 - Maximum time: 0.015 s (15.00 ms)
- Test Results:**
 - 5 out of 5 shown test case(s) passed
 - 5 out of 5 hidden test case(s) passed
- Test Case 1 Details:**
 - Duration: 15 ms
 - Status: PASS (green checkmark)
 - Buttons: DEBUG, menu, and expand
 - Expected output:**
 - 4
 - 55 65 78 89
 - Aggregate Percentage: 71.75%
 - Grade: First Division
 - Actual output:**
 - 4
 - 55 65 78 89
 - Aggregate Percentage: 71.75%
 - Grade: First Division

1.2.2. Fibonacci Series

20:08

Write a Python program that uses recursion to print the first `n` terms of the Fibonacci series.

Input Format:

- A single integer `n` representing the number of terms to generate.

Output Format:

- A single line containing the first `n` terms of the Fibonacci sequence, separated by spaces.

'''

```
def fibonacci(n):
```

```
    if n<=1:
```

```
        return n
```

```
    return fibonacci(n-1) + fibonacci(n-2)
```

```

n = int(input())
for i in range(1, n + 1):
    print(fibonacci(i), end=" ")
'''

def fibonacci(n):
    if n<=1:
        return n

    return fibonacci(n-1) + fibonacci(n-2)

```

```

n = int(input())
for i in range(n):
    print(fibonacci(i), end=" ")

```

Average time
0.017 s
17.00 ms

Maximum time
0.045 s
45.00 ms

✓ 2 out of 2 shown test case(s) passed
✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 9 ms

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Expected output	Actual output
5 0 · 1 · 1 · 2 · 3 ·	5 0 · 1 · 1 · 2 · 3 ·

✓ Test case 2 6 ms

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1.2.3. Pattern - 1

12:40

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.